

A-02 : Part 1

$$\begin{aligned}
 1. \quad m(a+bX) &= a+b \cdot m(X) \\
 &= \frac{1}{N} \sum_{i=1}^N (a+bX_i) \\
 &= \frac{1}{N} \left(\sum_{i=1}^N a + b \sum_{i=1}^N X_i \right) \\
 &\quad \text{constant: average cancels out} \quad \text{scalar: factors out} \\
 &= \frac{1}{N} (Na) + b \left(\frac{1}{N} \sum_{i=1}^N X_i \right) \\
 &= a + b \cdot m(X) \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 2. \quad \text{cov}(X, a+bY) &= b \cdot \text{cov}(X, Y) \\
 &= \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (a + bY_i - m(a+bY)) \\
 &\quad * \text{ from Q1: } m(a+bY) = a + b m(Y) * \\
 &= \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) \cdot b (Y_i - m(Y)) \\
 &\quad \text{scalar: factors out} \\
 &= b \cdot \text{cov}(X, Y) \quad \checkmark
 \end{aligned}$$

$$3. \quad \text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X) \quad \text{AND} \quad \text{cov}(X, X) = s^2$$

$$\begin{aligned}
 \text{cov}(a+bX, a+bX) &= \frac{1}{N} \sum_{i=1}^N ((a+bX_i) - m(a+bX))^2 \\
 &= \frac{1}{N} \sum_{i=1}^N (b(X_i - m(X)))^2 \\
 &= b^2 \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2 \\
 &= b^2 \cdot \text{cov}(X, X) \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{cov}(X, X) &= \frac{1}{N} \sum_{i=1}^N (X_i - m(X)) (X_i - m(X)) \\
 &= \frac{1}{N} \sum_{i=1}^N (X_i - m(X))^2 \\
 &= s^2 \quad \text{by definition of variance} \quad \checkmark
 \end{aligned}$$

$$4. \quad \text{non-decreasing } g() \\ m(g(X)) = g(m(X)) ?$$

using counterexamples:

$$g(X) = X^3, \quad X \geq 0$$

$\uparrow$ non-decreasing, positive slope for domain $X \geq 0$

$$\text{using: } X = 1, X = 3$$

$$\begin{aligned}
 m(g(X)) &= m(g(1), g(3)) \\
 &= m(1, 27) \\
 &= 14
 \end{aligned}$$

$$g(m(X)) = g(2) = 8$$

$$14 \neq 8 \quad \checkmark$$

No, $g(m(X))$ does not always equal $m(g(X))$